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**UNITED STATES: A GOLDEN GROWTH
RATE AND CREDIT COUNTERPARTS
ANALYSIS**

Steve H. Hanke, John Greenwood, & Qiqin Sun

Johns Hopkins Institute for Applied Economics,
Global Health, and the Study of Business Enterprise



United States: A Golden Growth Rate and Credit Counterparts Analysis

By Steve H. Hanke, John Greenwood, and Qiqin Sun

About the Series

The *Studies in Applied Economics* series is under the general direction of Prof. Steve H. Hanke, Founder and Co-Director of the Johns Hopkins Institute for Applied Economics, Global Health and the Study of Business Enterprise (hanke@jhu.edu). The views expressed in each working paper are those of the authors and not necessarily those of the institutions' that the authors are affiliated with.

About the Authors

Steve H. Hanke is a Professor of Applied Economics and Founder & Co-Director of The Johns Hopkins Institute for Applied Economics, Global Health, and the Study of Business Enterprise in Baltimore. Prof. Hanke is a well-known currency reformer, and a currency and commodity trader. Prof. Hanke served on President Reagan's Council of Economic Advisers, has been an adviser to five foreign heads of state and five foreign cabinet ministers, and has held a cabinet-level rank as a state counselor in both Lithuania and Montenegro. He has been awarded seven honorary doctorate degrees and is an Honorary Professor at four foreign institutions. He was President of Toronto Trust Argentina in Buenos Aires in 1995, when it was the world's best-performing mutual fund. Currently, he serves as Chairman of the Supervisory Board of Advanced Metallurgical Group N.V. in Amsterdam. In 1998, he was named one of the twenty-five most influential people in the world by *World Trade Magazine*. In 2020, Prof. Hanke was named a Knight of the Order of the Flag.

John Greenwood was until December 2021 the Chief Economist of Invesco Ltd., a global asset management company. As editor of the Asian Monetary Monitor over the period 1977-96, he is widely credited as the designer of the restored currency board system in Hong Kong at the time of the currency crisis in 1983, a model for numerous subsequent currency board systems. Holding an MA and Honorary Doctorate from the University of Edinburgh, he is the author of "Hong Kong's

Link to the US Dollar – Origins and Evolution” (Hong Kong University Press, 2007 and 2022). He has been a Fellow at the Johns Hopkins Institute for Applied Economics, Global Health, and the Study of Business Enterprise since June 2016.

Qiqin Sun (qsun20@jhu.edu) is a sophomore at the Johns Hopkins University in Baltimore, pursuing majors in Economics, with a minor in applied maths & statistics. He contributed to this paper as Research Assistant of the Monetary Analysis Team at the Institute for Applied Economics, Global Health, and the Study of Business Enterprise in Spring 2023. He will graduate in May 2025.

Abstract

This paper presents a Golden Growth Rate (GGR) analysis of the United States for the period 2010 through 2019. A credit counterpart analysis for the period 2009 through 2022 is also presented with the purpose of analyzing post-financial-crisis monetary growth.

SECTION 1: US Golden Growth Rate (GGR) Analysis¹

The GGR is the growth rate of the money supply that is consistent with hitting a defined inflation target. The deduction of the United States' GGR uses data from the 2010-2019 period to exclude the abnormal shocks of the COVID-19 pandemic on economic activities.

Calculation of the GGR features the income form of the Quantity Theory of Money (QTM): $MV=Py$, where M stands for the money supply; V stands for the average number of times the money stock is used in income transactions per unit time; P stands for the price level; y stands for real GDP. To operationalize the QTM, we can transform the income form into the percent-change form: $\Delta M + \Delta V = \Delta P + \Delta y$.²

One can calculate the GGR by rearranging the Percent-Change Form as: $\Delta M = \Delta P + \Delta y - \Delta V$, where ΔM represents the GGR in broad money supply, $M2$.³ In this expression, ΔP denotes the annual inflation target of the Federal Reserve;^{4,5} Δy is the average year-over-year (YoY) percentage change in real GDP in the United States. To obtain change in income velocity ΔV , we first divide nominal GDP figures by the year-end $M2$ money supply, $V = \frac{\text{End of year Nominal GDP}}{\text{End of year } M2}$. Then, we calculate discrete percentage changes in income velocity, $\Delta V(t) = \frac{V(t)-V(t-1)}{V(t-1)}$, and take the average of such changes as ΔV . Inflation rate data, real GDP growth, and nominal GDP are retrieved from the Economic Intelligence Unit.⁶

Then, we are able to plug in US economic statistics and calculate the GGR that is consistent with hitting the 2% inflation target. Between 2010 and 2019, the average US real GDP growth rate was 2.25%, while the average change in income velocity was -2.01%. Thus, based on the percent-change-form of the QTM, the GGR for $M2$ is $\Delta M = \Delta P + \Delta y - \Delta V = 2\% + 2.25\% - (-2.01\%) = 6.26\%$ per year. In reality, the average annual rate of change in $M2$ was 6.09% over the 2010-2019 period, slightly lower than the GGR. As a result, the average annual CPI inflation rate of the United States was 0.26 percentage points lower than the inflation target of 2%.

The graphs below show the time paths of US annual inflation rates compared to the inflation target and annual $M2$ growth rates compared to the GGR from 2010 through 2019.

¹ For treatment of the Quantity Theory of Money as it relates to Golden Growth Rates, see Greenwood-Hanke (2021) <https://onlinelibrary.wiley.com/doi/abs/10.1111/jacf.12479>

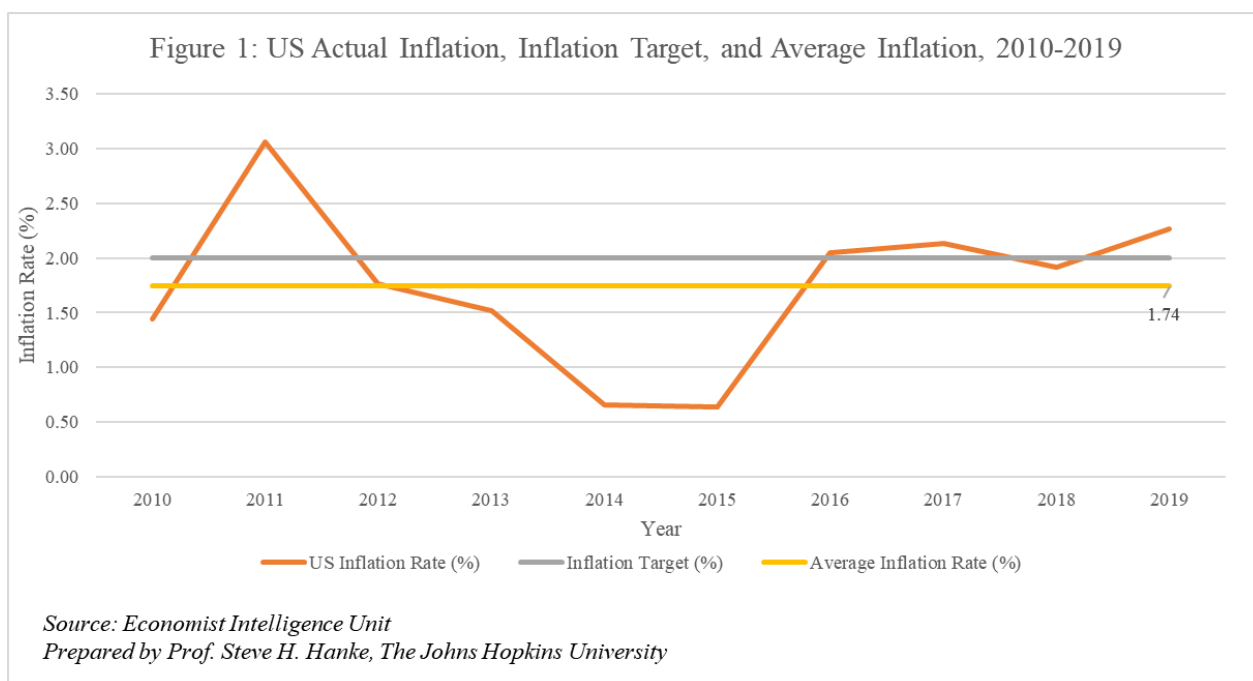
² For a derivation of the percent change form, see: Friedman 1987 <https://miltonfriedman.hoover.org/internal/media/dispatcher/214346/full>

³ Federal Reserve Economic Data. $M2$. <https://fred.stlouisfed.org/series/M2SL>

⁴ Board of Governors of the Federal Reserve System (2012). "Statement on Longer-Run Goals and Monetary Policy Strategy." https://www.federalreserve.gov/monetarypolicy/files/fomc_longerrungoals.pdf

⁵ The Federal Reserve announced its long-run inflation target in 2012 to be 2% on the Personal Consumption Expenditure Price Index (PCE). For simplicity, we apply the 2% target rate to CPI-based inflation statistics.

⁶ Economist Intelligence Unit (EIU). Data. <https://viewpoint.eiu.com/data/>



SECTION 2: Framework of Credit Counterpart (CC) Analysis – An Asset Approach⁷

Credit Counterpart analysis reveals the drivers of changes in broad money supply. The monetary liabilities constituting M2⁸ have counterparts on the asset side of commercial banks - bank lending, securities holdings, reserve assets, and residuals adjusting for discrepancies and omitted items. Expressed in terms of changes, we have:

Liability side (of central banks):

1. $\Delta \text{Broad Money} = \Delta \text{Bank deposits} + \Delta \text{Cash in circulation with the public, and}$

Asset side (of commercial banks):

2. $\Delta \text{Broad Money} = \Delta \text{Commercial Bank Lending} + \Delta \text{Securities (held by the banks)} + \Delta \text{Commercial bank reserves at the central bank} \pm \Delta \text{Others (net)}$

The second equation specifies the change in broad money supply as a sum of changes in major categories of assets held by commercial banks. Our analysis focuses on how market forces and the Fed's policies lead to changes in M2 through changes in commercial bank assets over time. The following section consists of a brief overview of US credit counterparts using annual data and a detailed analysis of how the Fed's operations influenced M2 through commercial banks' lending activities.

For our analysis, we use data from the nine asset accounts in the consolidated balance sheet of US commercial banks from 2009 through the end of 2022. This is released periodically in H.8 Tables by the Board of Governors of the Federal Reserve.⁹ All data are not seasonally adjusted and retrieved from the last weekly updates of each year or month from 2009 through 2022. Figure 3 shows the changes in M2 and major types of commercial bank assets in the United States from 2009 through 2022, using monthly data.

The accounts can be categorized into four types:

⁷ For logic of the Credit Counterpart Analysis, see: Hanke (2017). <https://www.e-elgar.com/shop/usd/money-in-the-great-recession-9781784717827.html>

⁸ U.S M2 = (1) currency outside the U.S. Treasury, Federal Reserve Banks, and the vaults of depository institutions + (2) demand deposits at commercial banks (excluding those amounts held by depository institutions, the U.S. government, and foreign banks and official institutions) less cash items in the process of collection and Federal Reserve float + (3) other liquid deposits, consisting of other checkable deposits (or OCDs, which comprise negotiable order of withdrawal, or NOW, and automatic transfer service, or ATS, accounts at depository institutions, share draft accounts at credit unions, and demand deposits at thrift institutions) and savings deposits (including money market deposit accounts) + (4) small-denomination time deposits (time deposits in amounts of less than \$100,000) less individual retirement account (IRA) and Keogh balances at depository institutions + (5) balances in retail money market funds (MMFs) less IRA and Keogh balances at MMFs.

Board of Governors of the Federal Reserve System. Money Stock Measures – H.6

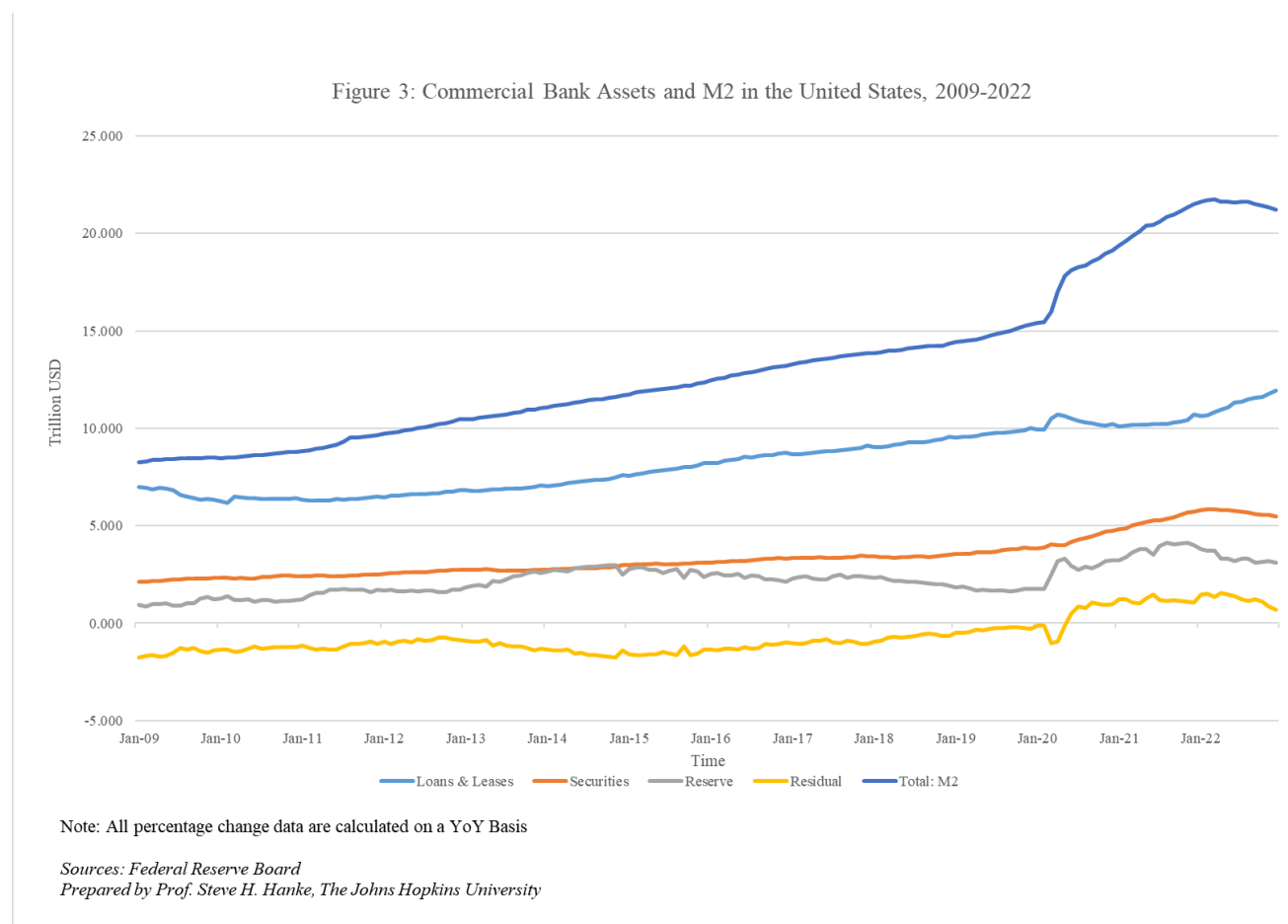
<https://www.federalreserve.gov/releases/h6/current/default.htm>

⁹ Board of Governors of the Federal Reserve System. Assets and Liabilities of Commercial Banks in the United States - H.8..

<https://www.federalreserve.gov/releases/h8>

- (1) *Loans and Leases* – sum of real estate loans, commercial and industrial loans, consumer loans, all other loans and leases, and loans to commercial banks minus allowance.
- (2) *Securities* – treasury and agency securities, plus other securities.
- (3) *Reserve and Cash* – the cash assets held by commercial banks including deposits at the Fed.
- (4) *Residuals* – M2 less the sum of the three categories of assets identified above.

The credit counterpart analysis breaks down the policies of the Federal Reserve into two major periods - the pre-pandemic period (2009-2019) and the pandemic period (2020-2022). Each period is made up of two stages - monetary easing at first and subsequent tightening. Shortly recapping the policy tools of the Fed is necessary before jumping into analysis.



The following is a summary of the Fed's monetary policies used in the pre-pandemic period:

- **Quantitative Easing (QE):** QE is the monetary policy tool whereby the central bank purchases a large, planned amount of government bonds or other securities to offer liquidity in the financial market. In QE1 (November 2008), the Fed initially purchased \$600 billion of mortgage-backed securities and then ratcheted up the scale and scope of the assets that it purchased. In QE2 (November 2010), the Fed aimed to buy \$600 billion of Treasury securities by the end of June 2011. In QE3 (September 2012), the Federal Reserve planned to additionally conduct \$40 billion in open-ended purchases of

mortgage-backed securities per month, and the volume of purchase was adjusted subsequently with regard to economic conditions.¹⁰

- **Changes in the Fed's Operating Procedures:** To establish the federal funds market as the principal target of monetary policies, the Fed started to pay interest on banks' reserve deposits at the Fed in 2008.¹¹ It also created an overnight reverse repurchase agreement (ON RRP) facility in 2014.¹² Simultaneously, the Fed allowed the federal funds target to rise above the zero bound for the first time after 2008.¹³
- **Quantitative Tightening (QT):** QT is the reverse action of QE. After 2015, the Fed aimed to stabilize its balance sheet and formally initiated a QT program in 2018 by retiring some of its expired debts. The size of its balance sheet shrank rapidly in 2018 and 2019, but such pressure on economic activities forced QT to stop after one year.¹⁴

During the COVID period, the Fed again pursued expansionary monetary policies, and with a typical lag, inflation followed. The following are a few policy tools the Fed used to overstimulate and then cool down the economy from 2020 through 2023:¹⁵

- **QE:** The Fed set its rate of purchases to \$80 billion per month in Treasuries and \$40 billion in residential and commercial mortgage-backed securities in March 2020. The pace of asset purchase remained intact until November 2021. It also purchased commercial paper as part of the asset purchase program.
- **Federal Funds Rate:** The Fed lowered the federal funds target to 0-0.25% in March 2020. However, as inflation hit high points afterwards, the Fed decided to lift the target rate seven times consecutively in 2022, pulling it to 4.25%-4.5% at the end of the year.¹⁶
- **Direct Lending Programs:** In 2020, the Fed started to offer more direct loans to banks by cutting the discount rate from 2.25% to 0.25% to prevent bank panics. It also created programs that lent directly to securities firms, state and local governments, and major corporate employers.
- **Reserve Requirement:** In March 2020, the FOMC decided to lower the reserve requirement of commercial banks to 0%, which was confirmed formally in 2021.
- **Repo Facilities:** The Fed extended the lengths and amount of its repo facilities in the federal funds market and set up a permanent Standing Repo Facility to smooth money markets in July 2021.

¹⁰ Stephen D. Williamson (2017). "Quantitative Easing: How Well Does This Tool Work?"

<https://www.stlouisfed.org/publications/regional-economist/third-quarter-2017/quantitative-easing-how-well-does-this-tool-work>

¹¹ Board of Governors of the Federal Reserve System. Interest on Reserve Balances.

<https://www.federalreserve.gov/monetarypolicy/reserve-balances.htm>

¹² Board of Governors of the Federal Reserve System (2014). "Federal Reserve issues FOMC statement on policy normalization principles and plans." <https://www.federalreserve.gov/newsevents/pressreleases/monetary20140917c.htm>

¹³ Federal Reserve Economic Data. Federal Funds Target Range – Upper Limit. <https://fred.stlouisfed.org/series/DFEDTARU>

¹⁴ Kristie M. Engemann (2019). "What is Quantitative Tightening?"

<https://www.stlouisfed.org/open-vault/2019/july/what-is-quantitative-tightening>

¹⁵ Eric Millstein and David Wessel (2021), "What did the Fed do in response to the COVID-19 Crisis."

<https://www.brookings.edu/research/fed-response-to-covid19/>

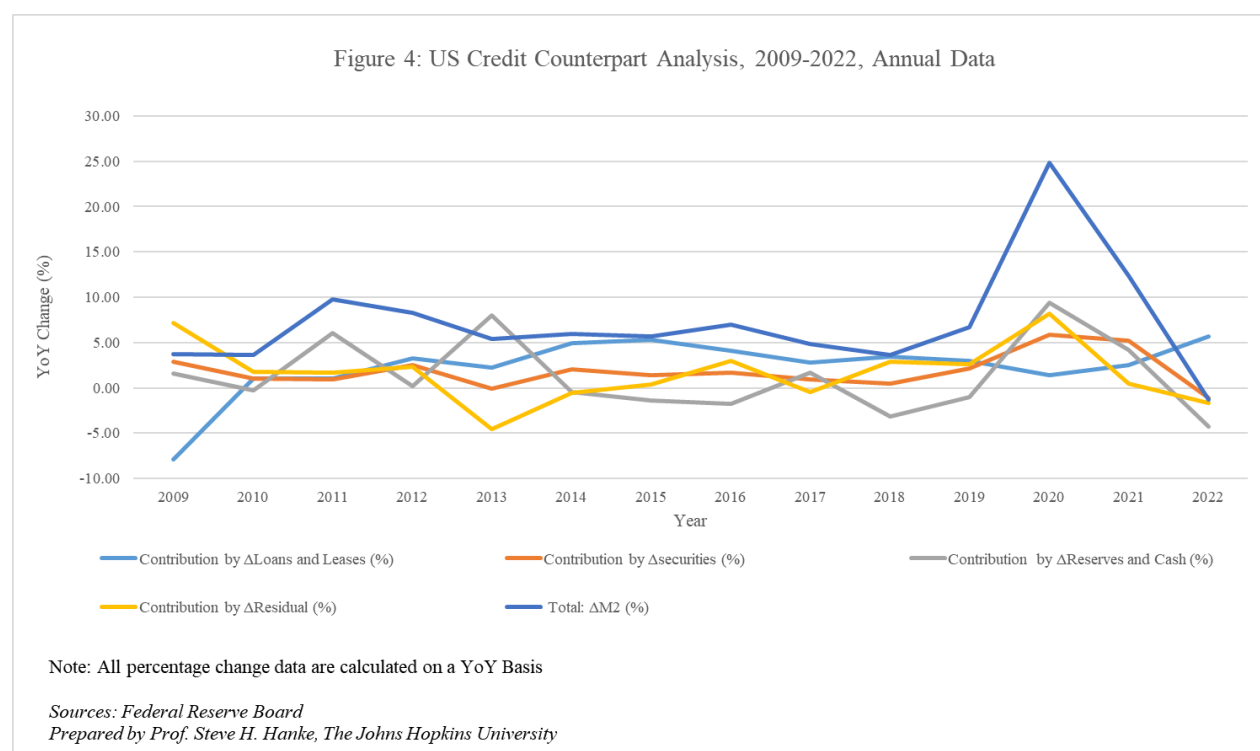
¹⁶ Board of Governors of the Federal Reserve System. Open Market Operations.

<https://www.federalreserve.gov/monetarypolicy/openmarket.htm>

- **QT:** As CPI inflation became troublingly high in 2021, reaching 9.1% YoY in July, the Fed halted reinvestments in maturing Treasury bills and in maturing MBS each month. The purchases were subject to \$30 billion and \$17.5 billion caps, respectively.¹⁷

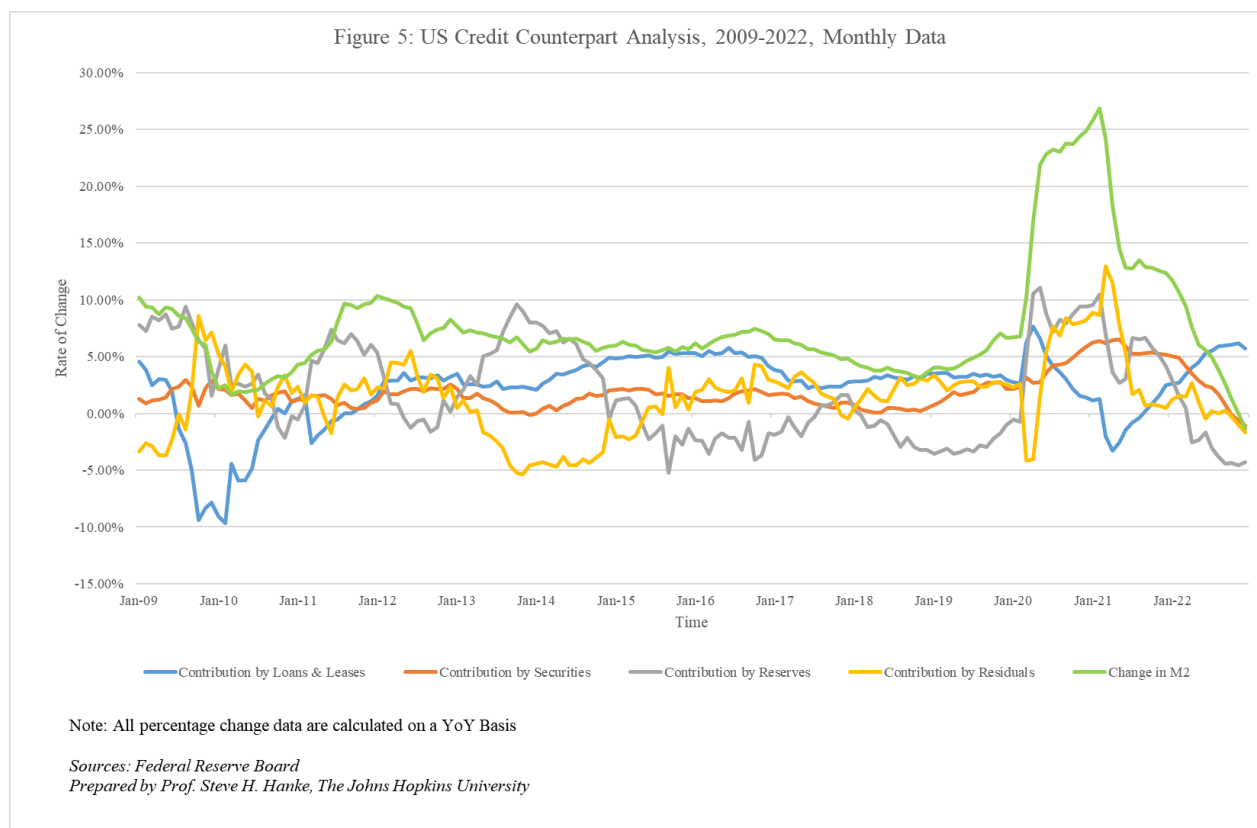
To give an insight into the impacts of the Fed's operations, we can distinguish the contribution of changes in each category of bank assets to the changes in M2. Figure 4 on the next page shows the contribution of bank assets to M2 growth using annual data. For example, in January 2016, where the contribution of reserves was -2.37 percentage points, the contribution of loans and leases was 5.31 points, and the contribution of residuals was 1.87 percentage points, which added up to 6.17% - the YoY growth rate in M2.

There are a few general takeaways from Figure 4. One important observation is that contributions of reserves were significant only after the Fed conducted monetary expansions, namely in 2010-2011, 2013, and 2020-2021 periods. Another notable observation is that loans and leases were more important drivers of monetary expansions compared to securities, but changes in loans and leases were also more volatile than those of securities. Lastly, contributions of residuals were usually insignificant but moved drastically in 2009, 2013 and 2020 in response to the Fed's monetary expansions.



¹⁷ Tim Sablik (2022). "The Fed Is Shrinking Its Balance Sheet. What Does That Mean?" https://www.richmondfed.org/publications/research/econ_focus/2022/q3_federal_reserve

We now provide more detail with monthly data. As elaborated before, we can divide the period of 2009-2022 into three periods: 2009-2015, 2016-2019, 2020-2022. The monthly YoY contributions of commercial bank assets are shown in Figure 5.¹⁸



- (1) **2009-2015:** As seen in Figure 5, the effects of the three QEs were significant on reserves held by commercial banks, resulting in three peaks in their contributions to the growth of M2. However, QE1 did not result in the stimulation of bank lending, despite the provision of plentiful reserves. Consequently, the Fed came up with QE2, which stabilized the contribution of loans and leases in positive ranges in 2011, and QE3 to boost the recovery of YoY increases in lending activities. Eventually, the contribution of loans and leases recovered in 2012. Simultaneously, the banks were not willing to increase securities holdings drastically due to the slow recovery of the economy. The contribution of residuals was very positive before QE2 but declined sharply later on. Overall, growth in M2 was moderate in all years aside from 2010, with most of the monetary expansion being explained by the growth in commercial banks' reserves, but not credit expansions.
- (2) **2016-2019:** As economic activities went back to normal levels, the Fed started to stabilize the size of its balance sheet after 2015 and retired parts of expired securities after 2018. As a result, the contribution of reserves to M2 was mostly negative during the period. Under

¹⁸ Data of Allowance for loans and leases is not available before July 2009, so there can be biases in YoY contribution margins before July 2010.

tighter conditions in the money market, the commercial banks decelerated credit expansions by reducing their lending and securities holdings. As a result of the fact that QT had created a liquidity crisis, the Fed quickly abandoned its QT scheme in September 2019, and injected \$75 billion into the reverse repo market. With insignificant levels of residuals, the M2 growth rate underwent slight decreases compared to the previous period mainly because of slower expansions in lending and negative changes in reserves.

- (3) **2020-2022:** During the pandemic, the Fed cut the federal funds rate sharply between early 2020 and mid-2021 and conducted QE. The contribution of reserves surged up to 10% in April and May 2020 and remained high throughout most of 2021. Meanwhile, the commercial banks refrained from increasing their lending activities drastically as a recession occurred. Instead, residuals rose quickly along with securities holdings. All credit counterparts, aside from lending, supported the extremely high YoY growth rate in M2 in 2020 and 2021.

To counter the high inflation, resulting from excessive monetary growth in 2020-21, the Fed started to lift federal funds targets and introduced QT. In consequence, the contribution of reserves to the money supply abruptly fell in just a few months. The contribution of securities declined, and the contribution of lending increased. The contribution of residuals also plummeted almost instantly after the Fed put forth the policies. Despite the acceleration in lending, the overall result of sharp decreases in the contribution margins of reserves, securities, and residuals was a plummeting rate of YoY growth in M2 from above 10% in December 2021 to negative levels in December 2022.

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